bank-cua

Record-once / replay-many computer use for legacy back-office banking apps

Design report & executive summary · Surya Prabhav Gurram

Executive summary

Banks and credit unions run a long tail of legacy applications with no API — the only way in is to drive the UI like a human operator. **bank-cua** lets an LLM figure out a task *once*, compiles that run into a typed, reviewable **capability artifact**, and then **replays it deterministically with no model in the loop** — the cheap, reliable path an AI agent triggers in production.

> The model discovers → the artifact becomes a reusable capability → deterministic replay is how the agent invokes it in production.

What was built (end-to-end vertical slice)

Discovery — goal-driven observe→decide→act loop drives a real, live UI.

Replay — deterministic, no-LLM execution with stable locators + checkpoints.

Safety — allowlist, irreversible-action gating, secret/PII redaction.

Cross-tenant — one artifact reused across a rebranded 2nd tenant via a small override.

Confidence — multi-run stability signal + draft→approved gate; bounded assisted recovery.

Artifact — typed, versioned, parameterised capability contract (JSON).

Error taxonomy — business-outcome vs. recoverable vs. hard-failure.

Escalation — human takes control of the *same live session* over CDP.

Agent API + codegen — callable-by-name HTTP endpoints; artifact → runnable Playwright.

Proven, not described — from `evidence/`

Scenario	Result
Member lookup, valid member (12345)	**SUCCESS** · outputs `member_name`, `savings_balance=421355` (cents)
Member 00000 / 99999	**BUSINESS_OUTCOME** · `MEMBER_NOT_FOUND` / `PERMISSION_DENIED` (not a crash)
Unexpected maintenance interstitial	**RECOVERED** · auto-dismissed, run still succeeds
Session timeout injected	**HARD_FAILURE** · `SESSION_TIMEOUT` surfaced w/ screenshot + DOM
Sub-account creation (irreversible)	**HANDOFF** · paused → operator confirmed on same session → resumed to SUCCESS
Same artifact on a 2nd tenant (rebranded labels)	**CROSS-TENANT** · clean with a ~4-line override; still works via fallback (drift) without one
Replayed N times	**STABILITY** · pass-rate signal feeding the draft→approved gate

Design stance

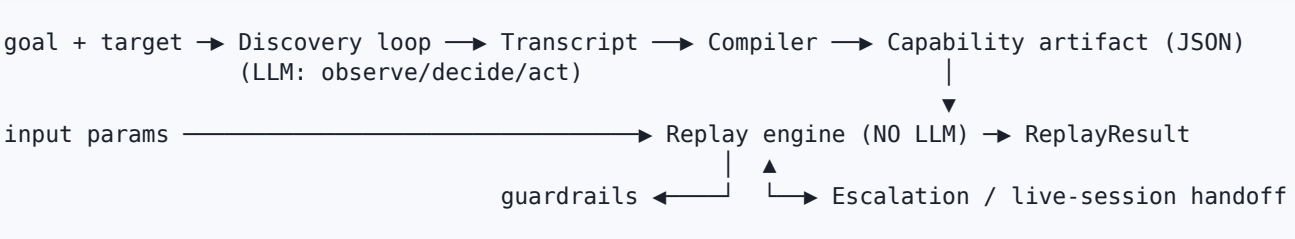
Go deep on the load-bearing pieces — the artifact schema, deterministic replay plus its error contract, and the safety/escalation model — and keep everything else thin at a clean seam. The `Surface` abstraction and the flow-vs-tenant-binding split in `Target` keep legacy-web, desktop, and multi-tenant reuse out of the core's way without building that plumbing prematurely. Stack: Python + Playwright, Pydantic v2 schema, a legacy-style Flask mock as the proxy target.

Full source, artifacts, and evidence in the accompanying public repository. 87 automated tests pass; a repo-wide sweep finds no secret in any artifact, log, or transcript.

Design report

1 · Architecture

The system is one Python package with sharp internal seams and a single dependency direction: everything speaks the **schema** and the **surface** interface; the only places that speak Playwright are the web surface and the operator's CDP client (the handoff).



- **Surface (perceive/act) vs. the recorded flow.** The schema, discovery loop, and replay engine import only this interface. A legacy-web frame driver or a desktop accessibility driver is a new `Surface` and touches nothing else.
- **Discovery and replay share the same targeting primitives.** Discovery synthesises a Locator from the element it chose and acts *through* it, so what is recorded is exactly what is exercised — it already worked once.
- **The model never writes selectors.** It picks elements by an integer `ref`; the system computes the robust, multi-strategy Locator. Locator quality is a property of code, not prompt luck.
- **The compiler is the record→capability boundary** — parameterising, synthesising checkpoints, attaching the vendor condition library, referencing (not inlining) a redacted transcript.
- **Single process, synchronous, file-backed.** No queue, no services; the parts that would scale (artifact-as-data, catalog, handoff inbox) are already serialisable and stateless. Simplicity is deliberate.

2 · Artifact schema

A `CapabilityArtifact` is a **contract**, not a step dump: a calling agent should understand what it needs and returns without reading the steps.

- **Contract-first** — typed `inputs` (with `required`, `sensitive`, `example`), typed `outputs`, and a top-level `success` checkpoint. The catalog turns these into an agent function-calling manifest.
- **Robust targeting as data** — every control is a frame-aware `Locator` with an **ordered candidate list, most-stable-first** (role+name → label → placeholder → text → CSS → XPath), each with human `reasoning`. Read-only values get **label-relative XPath** (`//tr[td="Savings"]/td[last()]`) that survives table reflow.
- **The error taxonomy lives in the schema** — `KnownCondition` records a detector, a class, and an optional recovery, as reviewable data.
- **Safety in the schema** — each step carries a `RiskClass` and `requires_confirmation`; the compiler flags create/confirm/submit as risky.
- **Reuse-ready target** — `Target` separates the shared flow from the per-tenant binding (`base_url`, `tenant_id`, `vendor_product`, `version`).

- **Provenance, decoupled from the transcript** — semver, approval state, `recorded_by`, and a transcript reference + sha256; the raw transcript is stored separately and redacted, never inlined.

3 · Determinism & error handling

Determinism. Replay runs with no model: it resolves each control via the ordered candidates, waits for declared conditions instead of sleeping, and asserts a **per-step checkpoint** after acting before proceeding. Parameters are substituted into URLs, values, and checkpoints at run time.

The runtime-error contract is the core. After every step (and on any action failure) replay runs the artifact's condition detectors and acts on the first match:

- `business_outcome` stop and return a code the caller wants — "no such member" / "permission denied" are *answers*, never crashes.
- `recoverable` run the declared recovery (dismiss interstitial, reload transient 500) up to its budget, then re-scan and continue; each recovery is recorded.
- `hard_failure` stop with a structured error: step, expected, observed, plus screenshot + redacted DOM snapshot.

A checkpoint that fails with no matching condition is itself a hard failure with full evidence — unrecognised trouble never silently proceeds. UI drift (secondary) is absorbed by the ordered locator fallbacks and by checkpoints failing loudly; a stability signal and an approval gate are the hooks to catch drift before unattended production.

4 · Heterogeneity & multi-tenant

Other surfaces. The seam is `Surface`. The schema speaks roles, names, labels, and frames — vocabulary a desktop **accessibility tree** exposes just as a browser does — so the format assumes no DOM. Legacy web is the same surface leaning on `frame_path` and structural fallbacks; desktop is a new `Surface` over an OS accessibility API or screenshot+coordinates (the schema already has a coordinate locator kind). Perception/action change; the flow, compiler, replay, error contract, and safety do not.

Multi-tenant reuse. `Target` splits the shared flow from the per-tenant binding, so hundreds of tenants on one vendor product share a single artifact and differ only by binding + optional overrides. Two mechanisms make this real: **vendor-level condition libraries** (how "Corebank" signals a locked account is curated once and inherited by all its tenants), and **semantic-first locators** that survive per-tenant theming better than structural paths. Drift is detected by watching checkpoint failures and which locator candidate resolved (both logged); a per-tenant stability score gates unattended use. Canonicalising routes (`/item/12345 → /item/:id`) extends the existing URL parameteriser. **Built and demonstrated:** a rebranded second tenant variant (`MOCKBANK_VARIANT=summit`, "Member Number"/"Find"/"Log In"); the member-lookup artifact recorded on `demo-cu` runs against `summit-cu` — clean with a ~4-line override, and still succeeding via structural fallbacks (with drift signals) without one. Only the multi-tenant *plumbing* (registry, scheduling) is left unbuilt, per the brief.

5 · Escalation & handoff

Detecting "stuck." Discovery escalates on no-progress over N steps, an unresolvable target, a policy block, or a model `escalate`. Replay escalates on a policy-gated irreversible step or an unrecoverable condition.

Routing with context. An `InterventionRequest` carries the capability/goal, current step, live URL, screenshot + DOM, reason, a **control token** (`agent` / `operator`), and the **CDP endpoint** of the live session, to a file-backed inbox; the automation blocks on it (and, unattended, times out and aborts cleanly, preserving the request for triage).

Taking control of the same live session. The browser is launched with a remote-debugging port, so the operator attaches to the **same Chromium** via `connect_over_cdp` and drives the **same page** — cookies, half-filled forms, and URL all preserved. Human actions are recorded; on resolve the token returns to `agent`; replay resumes, or skips execution if the human performed the gated step (so an irreversible action is never double-executed). The full round trip runs in evidence and via the `operator` CLI. The control-transfer *model* is real; the operator *console* is a CLI stand-in for a co-browsing UI, which is out of scope.

6 · Safety

Guardrails run on **every action, before it happens, in both discovery and replay**; a violation raises, never warns-and-continues.

- **Layered allowlist** — a global policy (URL patterns, action types, an explicit denylist) plus the artifact's own allowlist; a URL must satisfy both, else it is blocked (fail closed).
- **Irreversibility, treated conservatively** — reversible actions are safe; create/confirm/submit/delete are risky and **blocked unless explicitly approved**, or gated behind human confirmation (an escalation when unattended). Refusing to create a sub-account is far cheaper than creating one by mistake.
- **Never persist secrets or regulated data** — sensitive inputs are stored as parameter *names*, never values; redaction is name-based + pattern-based (SSN, card, email) across logs, transcripts, and DOM snapshots. The observation indexer never captures a control's typed value, so secrets never reach the model prompt in the first place.

Limits. Screenshots can still show on-screen PII (treated as sensitive evidence today; masked capture next). The allowlist is URL/action-shaped, not semantic — value-level policy (amount limits, dual control) is the next layer. Risk classification is a keyword+action heuristic and should become an explicit per-step review gate before approval.

7 · Cuts

Most stretch goals are **built**: agent-facing API (`serve`), code generation (`codegen`), confidence/approval (stability signal + draft→approved gate), bounded assisted recovery, and cross-tenant reuse with per-variant overrides + route canonicalisation. Left thin at clean seams: the **operator console** (CLI stand-in; the handoff mechanism is real); **desktop/legacy-web surfaces** (designed, not built — the `Surface` seam keeps the core uncommitted); **multi-tenant plumbing** (registry/scheduling — the scaling infrastructure the brief says to avoid); and the **committed discovery evidence is a genuine live run** via the Anthropic Messages API (`recorded_by: anthropic`), with a key-free offline reproduction via the bridge provider.

Next, in order: (1) a second `Surface` (desktop accessibility tree) to prove the seam on a genuinely non-DOM target; (2) value-level semantic policy (amount limits, dual control on money movement); (3) a real co-browsing console over the existing CDP seam; (4) an artifact auto-repair loop that proposes locator/override updates when drift crosses a threshold, gated by the approval workflow.